

24-502-0102: Artificial Intelligence and Machine Learning

Core/Elective: **Core**

Semester: **1**

Credits: **4**

Course Description

Machine learning is programming computers to optimize a performance criterion using example data or past experience. This course is to discuss many methods that have their bases in different fields: statistics, pattern recognition, neural networks, artificial intelligence, signal processing, control, and data mining. Major focus of the course is on the algorithms of machine learning to help students to get a handle on the ideas, and to master the relevant mathematics and statistics as well as the necessary programming and experimentation.

Course Outcomes (CO)

After the completion of the course, the students will be able to:

CO	Course Outcome Statement	Cognitive Levels
CO1	Understand and explain the different types of the learning process, and key ethical considerations.	Understand
CO2	Learn to effectively prepare data for machine learning models through data cleaning, feature selection, and dimensionality reduction.	Apply
CO3	Implement and interpret linear and non-linear regression models, while comparing various classification techniques including tree-based, kernel, and ensemble methods.	Apply
CO4	Gain practical knowledge in identifying data clusters using various algorithms and discovering hidden patterns through association rule learning.	Analyze
CO5	Understand the basic building blocks of neural networks, implement the backpropagation algorithm, and explore the concept of MDPs and Q-learning.	Apply

Mapping with Program Outcomes

	PO1	PO2	PO3	PO4	PSO1	PSO2	PSO3	PSO4
CO1	3	-	-	-	3	1	1	2
CO2	3	-	-	-	3	1	1	2
CO3	3	-	-	3	3	1	3	2
CO4	3	-	-	3	3	1	2	2
CO5	3	-	-	3	3	1	3	2

Course Content

1. Introduction to AI - What is AI? A Brief History of AI - Different types of AI - Applications of AI - Problem Solving Methods – Heuristics. Knowledge Representation and Reasoning - Planning and Decision-Making; Ethics and Societal Impact of AI.
2. Machine Learning Fundamentals - Concept of Machine Learning: Definition, applications, types of learning (supervised, unsupervised, reinforcement) - Hypothesis Spaces and Inductive Bias - Learning Process- Machine Learning Ethics and Bias. Data Preprocessing and Feature Engineering: Data Representation - Data Preprocessing - Features and Types - Dimensionality Reduction – Feature Identification - Feature selection – Feature extraction - Feature Importance-High dimensional data and Manifolds.
3. Regression and Classification - Regression: Linear Regression – Non-Linear regression – evaluation metrics for regression– Classification: Binary, multi-class, and multi-label classification – lazy learners - tree-based techniques - kernel-based techniques - probabilistic techniques - and ensemble techniques – evaluation metrics for classification.
4. Clustering and Rule Mining - Clustering: Partitioning based – hierarchical based – density based– grid-based – model based - Rule mining: Apriori algorithm, FB Growth - association rules. Outlier Detection - LOF.
5. Artificial Neural Networks and Reinforcement Learning -Neural Networks: McCulloch-Pitts neurons, Hebb's networks, Hopfield networks, Boltzmann machines, Perceptrons, multilayer perceptrons, backpropagation. Reinforcement Learning: Markov Decision Processes (MDPs), Q-learning.

References

1. Ethem Alpaydin, Introduction to Machine Learning, 3e, MIT Press, 2014
2. Tom M. Mitchell, Machine Learning, McGraw Hill Education; 1e, 2017
3. Stephen Marsland, Machine Learning, An Algorithmic Perspective, 2e, CRC Press, 2015
4. Giuseppe Bonaccorso, Machine Learning Algorithms, 1e, Packt Publishing Limited, 2017
5. Ethem Alpaydin, Machine Learning- The New AI, MIT Press, 1e, 2016
6. Andrew Ng, Machine Learning Yearning, ATG AI (Draft version), 1e, 2018
7. Rohit Singh, Tommi Jaakkola, and Ali Mohammad. 6.867 *Machine Learning*. Fall 2006. Massachusetts Institute of Technology: MIT OpenCourseWare, <https://ocw.mit.edu>
8. Andrew Ng, <https://www.coursera.org/learn/machine-learning>